



Once burned, twice shy: Repeat fires reduce seed availability and alter substrate constraints on *Picea mariana* regeneration

Carissa D. Brown*, Jill F. Johnstone

Department of Biology, University of Saskatchewan, 112 Science Place, Saskatoon, Saskatchewan, Canada S7N 5E2

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ABSTRACT

Widespread climate change is expected to lead to altered patterns of disturbance, thereby driving future ecosystem change. This interaction, which is often poorly recognized or understood, may be particularly important in the sub-arctic due to rapid climate change and frequent fire. Our objective was to investigate how an altered fire return interval can interrupt successional pathways in a serotinous boreal ecosystem. We conducted this research in black spruce (*Picea mariana* [Mill.] BSP.) forests on the northern margin of the species' distribution in Yukon Territory, Canada. We compared seed availability and seedling establishment of black spruce in stands of varying fire return interval using experimental manipulations within areas varying in their natural fire histories. Recruitment was drastically reduced following two closely timed fires, compared to stands burned under a typical fire return interval. However, recruitment was also limited in mature forest stands. On-site germination experiments demonstrated that black spruce recruitment was limited by seed availability after a short fire return interval, and by substrate quality in unburned stands. The vegetation of the boreal forest is thought to be highly resilient to climatic change, due in part to the adaptations that conifers have for post-fire regeneration. We show that shortened fire return intervals, through effects on seed availability, disrupt the normal sequence of post-fire recovery on seedbeds released by fire for colonisation. The results of this study provide strong empirical evidence that disturbance, although essential for stand renewal, may limit forest recovery or expansion when misaligned with reproductive cycles.

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1. Introduction

The Earth has entered a period of rapid climate change that will have broad impacts on the structure and function of ecological systems (Millennium Ecosystem Assessment, 2005). Air temperatures are increasing in northern boreal regions (Chapin et al., 2005). Climate change will impact ecosystems not only via direct temperature effects, but also through indirect effects caused by changes to disturbance regimes. Fire patterns in the boreal forest, such as fire size and severity, are largely driven by weather and local moisture conditions (Renkin and Despain, 1992; Bessie and Johnson, 1995; Podur and Martell, 2009). In the boreal forest of North America, warmer temperatures, particularly extreme summer air temperatures, have been associated with increases in wildfire activity (van Wagner, 1987; Lee, 1995; Skinner et al., 2002; Parisien et al., 2011).

Abbreviations: MF, mature forest; LI, long-interval burned; SI, short-interval burned; GLM, generalised linear model; ZAP, zero-altered model with Poisson distribution; ZANB, zero-altered model with negative binomial distribution.

* Corresponding author. Present address: Département de Biologie, Université de Sherbrooke, Sherbrooke, Québec, Canada J1K 2R1. Tel.: +1 819 821 8000x66001; fax: +1 819 821 8049.

E-mail addresses: carissa.brown@usask.ca (C.D. Brown), jill.johnstone@usask.ca (J.F. Johnstone).

Consequently, although climate–fire interactions are complex, models consistently predict that fire activity will increase in the boreal forest due to climate change (Balshi et al., 2009; Flannigan et al., 2009). Factors other than climate, such as human activity and physiography, can also affect the fire regime, potentially causing regional declines in fire activity (Schulte et al., 2005; Wallenius, 2011). Nevertheless, climate-driven increases in fire activity are expected to dominate in wide regions of the boreal forest (Balshi et al., 2009; Flannigan et al., 2009). Consistent with this prediction, increases in fire frequency and area burned have already been observed in boreal forests of northwestern North America (Kasischke and Turetsky, 2006; Flannigan et al., 2009; Kasischke et al., 2010). Black spruce (*Picea mariana* [Mill.] BSP.)-dominated stands are common in this region and burn with a high frequency (Vioreck, 1983; Cumming, 2001), and are thus likely to be impacted early by a change in fire regime.

Disturbance characteristics exert a strong control over ecosystem structure and function, and the consequences of changing disturbance regimes for ecosystems and communities are a priority for global change research (Turner, 2010). Species that have adapted to regenerate with a specific disturbance regime are particularly sensitive to disruptions to that regime (Lamont et al., 1991). Serotinous species have a regeneration strategy that generates pulses of

regeneration via the release of large quantities of seed following a fire (Lamont et al., 1991; Lamont and Enright, 2000). A disruption of the historic fire regime experienced by a species may reduce the advantage of a serotinous regeneration strategy, particularly if a fire occurs before the species has reached reproductive maturity. Consequently, changes in the frequency of disturbance can disrupt dominance hierarchies even in the absence of additional environmental change (Paine et al., 1998; Bond et al., 2005). Constraints on succession caused by a disruption to the fire regime have been detected in serotinous systems globally (Keeley et al., 1999; Schoennagel et al., 2003; Johnstone and Chapin, 2006b; Westerling et al., 2006). In the boreal forest, fire is the main initiator of secondary succession (Payette, 1992) and several of the dominant conifers have regeneration strategies that rely on cone serotiny (Greene et al., 1999).

Black spruce is an example of a dominant boreal conifer that maintains seeds in an aerial seed bank of semi-serotinous cones for dispersal after fire (Greene et al., 1999). The individuals that establish during the first 5–10 years following fire, when recruitment and nutrient availability are at their highest, make up the majority of the stand that burns during the next fire (Johnson and Fryer, 1989). It can take two to three decades for black spruce to reach reproductive maturity and have viable seed available for post-fire self replacement (Burns and Honkala, 1990). In the western boreal forest, the interval between fires has historically been 80–150 years (Viereck, 1983; Larsen, 1997), by which time black spruce have reached reproductive maturity and produced enough seed to self replace after fire. Post-fire regeneration not only depends on seed availability, but also seed bed quality. Soil temperature and moisture, soil organic horizon thickness, and understory competition are all affected by fire history (Kasischke and Johnstone, 2005), and contribute to the ability of a seed to germinate and survive in a given location (Eriksson and Ehrlén, 1992; Greene et al., 1999). Changes in disturbance that affect seed availability or seed bed quality have the potential to alter patterns of post-fire recruitment and succession (Lloret et al., 2003; Johnstone and Chapin, 2006b; Caplat and Anand, 2009).

The objective of this study was to characterize the mechanisms by which a change in fire return interval may alter patterns of black spruce self replacement after fire. We hypothesized that black spruce recovery in stands experiencing a short fire return interval (14–15 years) would be severely reduced due to the lack of an aerial seed bank produced by the local, immature trees. We anticipated that these effects on seed availability would be more important than substrate quality in limiting recruitment after a short-interval fire. In contrast, we hypothesized that poor conditions for seed germination and survival would limit seedling recruitment in unburned stands. The natural microtopography of our study area also led us to hypothesize that earth hummocks, which tend to have less available moisture, warmer soils, and thinner soil organic horizons than the adjacent hollows (Black and Bliss, 1980; Kokelj et al., 2007), would have more black spruce recruitment than the colder, wetter hollows. We tested these hypotheses by documenting patterns of seed availability and seedling recruitment in stands of varying fire history. Results provide information on the relative importance of disturbance factors that constrain black spruce recruitment, including fire frequency and fire effects on substrate conditions. This study addresses an important gap in our understanding of the consequences of changes in fire regime to successional dynamics in fire-prone forests. Our research provides strong empirical evidence of ecological mechanisms that may drive rapid changes in forest composition in a region currently experiencing climatic warming (Chapin et al., 2005) and associated increases in fire activity (Kasischke and Turetsky, 2006).

2. Materials and methods

2.1. Study area

We established a field experiment in a natural mosaic of stands differing in recent fire history. Study sites were located in subarctic forests dominated by black spruce near Eagle Plains in northern Yukon Territory, Canada (66°06'56"N, 137°16'11"W). The region is characterized by rolling hills with fine grained loess soils over weathered bedrock (Smith et al., 2004). The surface microtopography is dominated by small-scale earth hummocks formed by differential freezing and thawing of soil (Kokelj et al., 2007). Soils are underlain with continuous permafrost with a seasonal thaw depth of less than 1 m (Tarnocai et al., 1993). Eagle Plains has a continental climate with a mean annual air temperature of -7.5°C , and monthly mean temperatures of -25°C in January and 13°C in July (Environment Canada, 1993). The mean annual precipitation is 400 mm, about half of which falls as rain in the summer (Environment Canada, 1993). Vegetation in the region is dominated by black spruce in the forest canopy with an understory of evergreen shrubs, willows (*Salix* spp.), and *Sphagnum* and feather mosses (Russell et al., 1992).

The fire history mosaic at Eagle Plains allowed us to sample mature unburned forests and forests that burned after long (~ 94 year) or short (14–15 year) fire-free intervals, all within close proximity to one another (Fig. 1). In 1990 and 1991, two adjacent regions of Eagle Plains were burned during lightning-ignited fires (1990: 33,500 ha and 1991: 4800 ha). A third fire burned the area in 2005 (69,000 ha), partially overlapping both the 1990 and 1991 burns (see Brown and Johnstone, 2011 for detailed fire information). We sampled sites representing three combinations of fire history and post-fire forest recovery: (1) mature, late-successional forest representing recovery from a long fire-free interval (previous fire ~ 77 years ago; MF), (2) early successional stands recovering from a long-interval burn (fire in 2005 following ~ 94 years fire-free; LI), and (3) early successional stands recovering from a short-interval burn (fire in both 1990/91 and 2005; SI).

2.2. Field measurements

Field observations and experimental manipulations were conducted within areas that varied in their recent fire histories. Eight 30 m $\times$ 30 m sites were selectively placed within each fire history class in an attempt to capture the natural variation of stand characteristics in each burn. Within each fire history class described above (MF, LI, and SI), we used seed traps to estimate seed availability and seeded plots in a two-factor design (fire history $\times$ microtopography) to test for seed-bed effects. Each site contained three east–west transects composed of seed traps and seeded plots.

Six seed traps were positioned randomly along one 30 m east–west transect in each site. Seed traps consisted of a 50 cm $\times$ 25 cm $\times$ 5.7 cm deep plastic greenhouse tray with a piece of plastic artificial grass fastened to the bottom, to reduce wind scouring of trap contents (Johnstone et al., 2009). Each seed trap covered 0.125 m² of ground, thus the six traps sampled a 0.75 m² area in each experimental site. Seed traps were emptied as early as possible in June after snowmelt in 2008 and 2009 in order to prevent the seeds from germinating in the traps. A second annual collection of seeds was made at the end of each summer, in August 2008 and 2009. The final seed trap collection occurred in mid-July 2010.

We measured viability of black spruce seed collected in the seed traps using laboratory germination trials (Leadem et al., 1997). Germination trials were conducted during the winter months (December–February) following each field season. Contents from

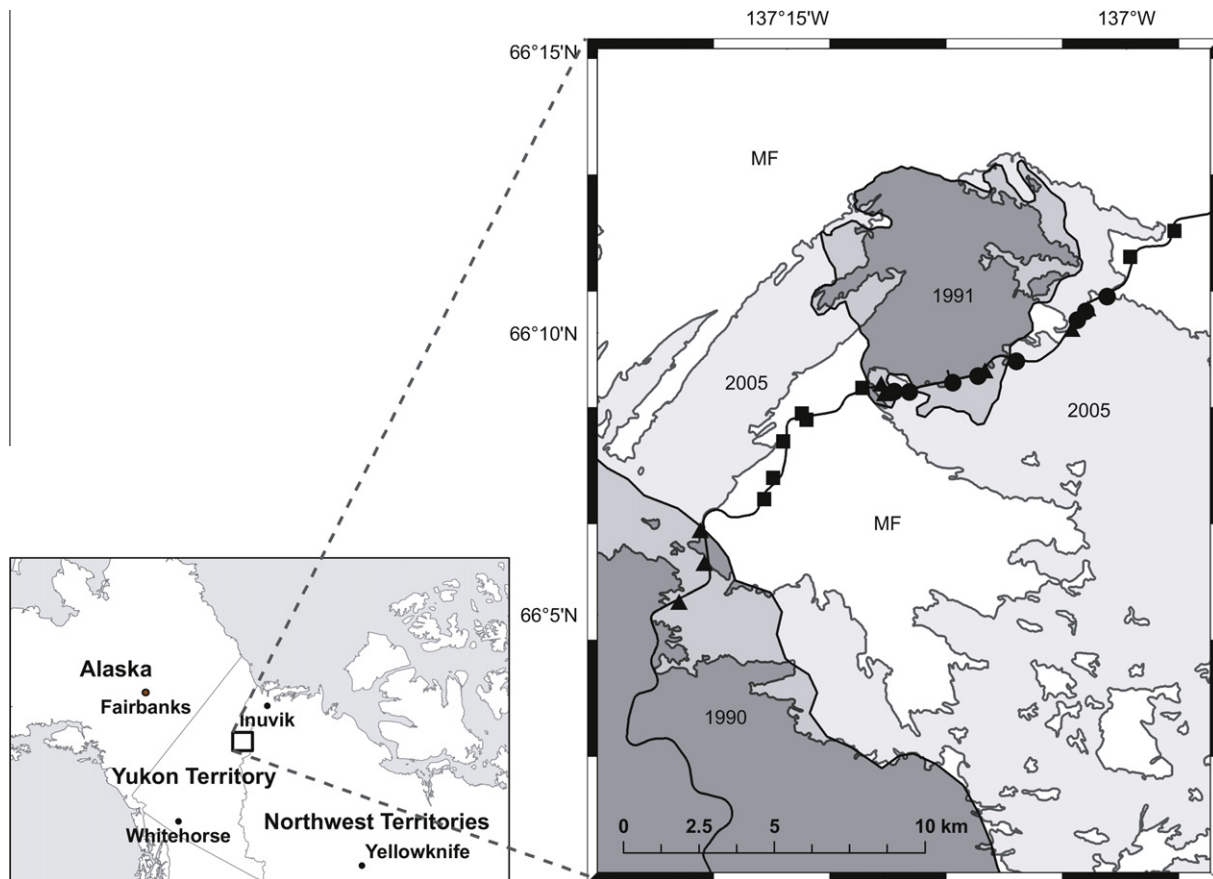


Fig. 1. Study area in northern Yukon, Canada, with areas burned in 1990/1991 (dark grey), 2005 (light grey), and the 1990s and 2005 fire overlap (medium grey) indicated. Mature forest (MF) is indicated in white. Study sites ($n = 8$ per fire history) were established in mature, unburned forest (squares), early successional long-interval burns (circles), and early successional short-interval burns (triangles). Modified from Brown and Johnstone (2011).

the six traps per site were pooled into a single sample and hand sorted to separate black spruce seed from other debris. The sorted samples were placed on moist filter paper in a Petri dish (9 cm diameter) at the beginning of the germination trial and were watered once per day with deionised water. Each site was tested in a separate dish, except where samples had more than 100 seeds and were divided into multiple dishes. Seeds were allowed to germinate for 28 days at 18 h of light per day at room temperature ($\sim 18^\circ\text{C}$), which mimicked mid-summer field conditions in the study area. Seeds were considered germinated when their hypocotyl grew to twice the length of the seed coat (Leadem et al., 1997).

Site differences in potential seedling recruitment were assessed with experimental applications of black spruce seed. The seed used in the trial was collected from subarctic black spruce forests in Tok, Alaska ($\sim 63^\circ 19' \text{N}$, $\sim 142^\circ 58' \text{W}$). In each site, 10 $0.5 \text{ m} \times 0.5 \text{ m}$ plots were seeded with 0.3 g (~ 250 seeds) of black spruce per plot. Seeded plots were positioned at even intervals along two 30 m transects and were stratified by microtopography, with five plots located on earth hummocks and five positioned in hollows. Each seeded plot was paired with an adjacent control plot to which no seeds were added. Seeds were evenly spread by hand across the surface of the forest floor in late summer (16–17 August, 2007 and 6–7 August, 2008) to allow for winter stratification of the seed prior to germination the following spring. Separate plots were seeded in 2007 and 2008 to distinguish annual effects on potential germination. Germinated seedlings in seeded and control plots were counted in June, July, and August in 2008 and 2009, and in July 2010. Because seedlings were monitored regularly, we were

able to detect new seedlings as they emerged and distinguish between recent seedlings and older recruits.

Seed availability typically increases with stand basal area and density (Greene and Johnson, 1999), thus these two variables were measured at each experimental site to characterize stand structure and seed source strength. Pre-fire stems were measured in burned sites whereas living stems were measured in mature stands. As nearly no wood biomass remained in the short-interval stands, and thus no seed source, we measured the distance from the edge of each short-interval stand to the nearest black spruce seed source. Seed source was defined as a stand of >20 living or recently burned (in the 2005 fire) mature black spruce. Distances were measured using a hand-held GPS (error $\pm 6 \text{ m}$).

2.3. Statistical analysis

Results from the seed trap collections and germination trials were summarized as total and viable annual seed rain per site. All individual seedlings that emerged within a plot during the study period were summed to calculate the total number of seedlings regardless of whether they survived past emergence. Individuals that survived at least 2 years were counted as 2-year recruits. We analysed data on total seedlings and 2-year recruits separately. For each of these two response variables, we pooled the plot-level seedling data within a site to obtain the total number of seedlings at the site-level. For all experimentally seeded plots, site-level counts of total seedlings and 2-year recruits were corrected by subtracting the total count from control plots within the same site. Correcting the seeded counts did not change the qualitative impacts of

fire history and allowed us to examine the seeding effect alone, without contributions from background natural recruitment.

Analyses of the seed trap data were conducted with generalised linear models (GLMs) using raw count data. Because of the structure of count data, we first constructed a model for each response variable from the seed rain data (total seed and viable seed) using GLMs with Poisson distributions, incorporating the natural logarithm as the link function (Zuur et al., 2009). We then ran the two models using GLMs with negative binomial error distributions. In both cases, a likelihood ratios test led us to select negative binomial models over Poisson distributions to account for over dispersion due to extra variation in the count data (Zuur et al., 2009). Seed rain models included fire history class as the predictor variable.

Seedling emergence data were also analysed with GLMs using raw count data. The two, individually modelled response variables (total seedlings and 2-year recruits) each had a large number of zeros observed, which is common in ecological data. We used zero-altered or hurdle models to deal with the excess number of zeros (Zuur et al., 2009). Zero-altered models consist of two parts. The data is first modelled as presence/absence of black spruce in a binomial model. Non-zero observations are then modelled with a truncated Poisson model (Zuur et al., 2009). As with the seed rain data, we first constructed a model for each response variable from the seeded plots (total seedlings and 2-year recruits) using zero-altered models with Poisson distributions (ZAPs), incorporating the natural logarithm as the link function (Zuur et al., 2009). We then ran the two models using zero-altered models with negative binomial error distributions (ZANBs). In both cases, a likelihood ratios test led us to select ZANBs over ZAPs to account for over dispersion due to extra variation in the count data (Zuur et al., 2009).

Both ZANB models had different predictor variables for the binomial model and count models, which we selected using standard model selection methods (Zuur et al., 2009). When modelling total seedlings, we included fire history, microtopography, seeding treatment, and an interaction term for fire history and seeding treatment as predictor variables. The count model of total seedlings included the same predictor variables, with the addition of an interaction term for fire history and microtopography. Our binomial model of 2-year recruits included fire history and microtopography as predictor variables. The count model of 2-year recruits included the same predictor variables, with the addition of microtopography and an interaction term for fire history and microtopography.

For all models, we assessed model fit using residual diagnostics (Zuur et al., 2009). All statistical analyses were conducted with R (R Development Core Team, 2009) using the “MASS” package for negative binomial GLMs (Venables and Ripley, 2002), the “pscl” package for zero-altered models (Zeileis et al., 2008; Jackman, 2010), and the “lme4” package for likelihood ratios tests (Zeileis and Hothorn, 2002).

3. Results

3.1. Site measurements

The patterns of stand basal area and distance to seed source in the study sites reflect the fire history of the study area. Due to the absence of woody biomass after two closely timed fires, SI sites had no standing dead black spruce and a mean distance of 71 ± 16 m to the nearest seed source. MF and LI sites, which had living and dead standing black spruce, respectively, had 0 m distance to the nearest seed source and basal areas of 7.9 ± 1.1 and 13.0 ± 1.9 m² ha⁻¹, respectively. Based on published equations relating stand basal area to seed production of black spruce (Greene and Johnson,

1999), these basal areas would be expected to represent an average of 184 and 296 seeds per m² in MF and LI stands, respectively.

3.2. Seed rain

Total ($X^2 = 87.00$, $df = 2$, $p < 0.001$) and viable ($X^2 = 13.35$, $df = 2$, $p = 0.001$) black spruce seed rain counts differed significantly among fire history classes (Table 1). This difference was due to SI sites, which had lower counts of both total ($z = -8.83$, $p < 0.001$) and viable ($z = -4.73$, $p < 0.001$) black spruce seed than MF and LI sites. These differences were apparent for seeds collected in all years (2008–2010). Total and viable black spruce seed rain did not differ between MF and LI sites in any year ($p > 0.05$; Table 1). The mean percent viability did not differ in MF, LI, and SI sites ($8.0 \pm 1.5\%$, $6.5 \pm 0.9\%$, and $7.2 \pm 4.8\%$, respectively; $F = 0.066$, $df = 2$, $p = 0.93$).

3.3. Total seedlings

Fire history, microtopography, and seeding treatment had significant effects on patterns of seedling emergence (Tables 2 and 3). The total number of seedlings was consistently lowest in MF sites and in control plots. Total seedlings in control plots, representing emergence from natural seed sources, was greatest in LI sites (Fig. 2a; Table 2). MF and SI sites had relatively little natural emergence over the 3 years of measurements. The binomial model, which was built upon presence/absence data, indicated that LI sites had significantly more plots with black spruce seedlings present than SI or MF sites, and black spruce seedlings were significantly more abundant in hollows than on hummocks. There was a significant interaction between fire history and seeding treatment. Adding seed to SI and MF sites changed the sign of the model estimate from negative to positive, meaning that seeded SI and MF sites tended to have black spruce seedlings present while control plots in those sites did not. This applied to SI sites seeded in 2007 and SI and MF sites seeded in 2008.

The count model indicated that in sites where black spruce seedlings were found, LI sites had significantly more seedlings than SI or MF sites, and black spruce seedlings were significantly more abundant in hollows than on hummocks. Adding black spruce seed resulted in significantly more seedlings than occurred due to natural recruitment (Fig. 2b). The effect of seeding on total seedlings was consistent for both 2007 and 2008 seeded plots (Tables 2 and 3). There was a significant interaction between fire history and microtopography. When black spruce seedlings were present in MF sites, more were found on earth hummocks than in hollows.

Table 1

Summary of total and viable seed rain across fire history classes (MF = mature forest, LI = long-interval burned, and SI = short-interval burned; $n = 8$ sites per fire history class). Data are means with standard error shown in parentheses. Values in bold indicate that the category differed significantly ($\alpha \leq 0.05$) from all others in the negative binomial count model.

Year collected	Total ^a seed rain per m ²			Viable ^b seed rain per m ²		
	MF	LI	SI	MF	LI	SI
2008	64.5 (19.1)	99.3 (15.4)	10.2 (3.7)	2.3 (0.6)	4.3 (1.1)	0.3 (0.2)
2009	49.7 (7.7)	52.7 (8.2)	6.9 (1.9)	5.4 (1.5)	5.3 (1.3)	0.9 (0.9)
2010	68.8 (15.9)	72.0 (12.6)	9.0 (2.4)	1.4 (0.4)	2.0 (0.6)	0.1 (0.1)
Average	61.0 (11.5)	74.7 (7.3)	8.7 (1.2)	5.3 (0.7)	6.4 (1.0)	1.9 (0.4)

^a Total number black spruce collected during each sample period.

^b Number of black spruce seed that germinated during each sample period.

Table 2
Summary of total seedlings and 2-year recruits counted in experimentally seeded plots in the three fire history classes (MF = mature forest, LI = long-interval burned, and SI = short-interval burned; *n* = 8 sites per fire history class). Data are also summarized by microtopography within each fire history class, with sites located either on earth hummocks or in hollows in hummocky terrain. Only total seedlings are presented for plots seeded in 2008, since 2 years had not passed between initial emergence after seeding and the last measurements. Counts in seeded plots were corrected by subtracting the counts from control plots within the same site. Mean data per m² are presented with standard error shown in parentheses.

	MF			LI			SI		
	Overall	Hollow	Hummock	Overall	Hollow	Hummock	Overall	Hollow	Hummock
<i>Total seedlings</i>									
Control	0.6 (0.2)	0.1 (0.1)	1.0 (0.3)	8.3 (1.6)	5.3 (0.8)	11.2 (2.7)	0.6 (0.2)	0.5 (0.3)	0.6 (0.3)
2007 Seeded	5.9 (2.5)	6.3 (3.1)	5.5 (4.3)	41.4 (7.2)	54 (11.6)	28.8 (6.6)	30.1 (11.9)	41.3 (22.3)	18.8 (8.5)
2008 Seeded	2.7 (1.0)	2.2 (1.1)	3.1 (1.7)	18.8 (4.0)	26 (6.7)	11.6 (3.0)	11.5 (4.9)	16.5 (9.7)	6.5 (2.1)
<i>2-year recruits</i>									
Control	0 (0)	0 (0)	0 (0)	4.2 (1.1)	2.5 (0.6)	5.9 (2)	0.1 (0.1)	0 (0)	0.2 (0.1)
2007 Seeded	2.4 (1.4)	1.9 (0.9)	2.9 (2.7)	19.5 (3.7)	25 (6.2)	13.9 (3.4)	15.3 (7.6)	24.1 (14.6)	6.4 (3.1)

Table 3
Summary of results from the zero-altered negative binomial model of total black spruce seedlings (*df* = 23). Experimental sites were located in three fire history classes (MF = mature forest, LI = long-interval burned, and SI = short-interval burned; *n* = 8 sites per fire history class), with sites stratified by two microtopography classes (earth hummocks and hollows in hummocky terrain) within each fire history class. Both models included fire history class, microtopography, seeding treatment, and the interaction between fire history class and seeding treatment as predictor variables. The count model also included the interaction between fire history class and microtopography as a predictor variable. For seeded plots, the models were based on counts corrected by subtracting counts from control plots within the same site. The intercept represents expected seedling counts in LI stands, hollows, and unseeded control plots. Values in bold indicate that a significant difference ($\alpha \leq 0.05$) was detected in the model.

Model term	Binomial model			Count model		
	Estimate	Standard error	z-Value	Estimate	Standard error	z-Value
Intercept	1.2223	0.2660	4.595	0.6889	0.2218	3.105
SI	−3.0806	0.4503	−6.842	−1.7466	0.7527	−2.321
MF	−3.0806	0.4503	−6.842	−1.9607	0.7414	−2.645
Hummock ^a	−0.8063	0.1781	−4.528	−0.5807	0.2080	−2.792
Seeded 2007 ^b	0.6158	0.3746	1.644	1.9412	0.2529	7.674
Seeded 2008 ^c	−0.2878	0.3400	−0.846	1.3600	0.2653	5.126
SI * hummock	–	–	–	0.1602	0.3676	0.436
MF * hummock	–	–	–	0.9723	0.4584	2.121
SI * seeded 2007	1.4892	0.5777	2.578	1.9083	0.7848	2.432
MF * seeded 2007	0.9445	0.5823	1.622	0.2272	0.8069	0.282
SI * seeded 2008	2.3928	0.5564	4.300	1.3465	0.7906	1.703
MF * seeded 2008	1.1926	0.5772	2.066	0.2774	0.8499	0.326

^a Plots located on hummock microtopography.
^b Plots seeded with black spruce in August, 2007.
^c Plots seeded with black spruce in August, 2008.

There was also a significant interaction between fire history and seeding treatment, as was detected in the binomial model. In the count model, however, this interaction was only significant for SI sites seeded in 2007.

3.4. 2-year recruits

Mortality of young seedlings resulted in fewer black spruce surviving to 2 years than originally emerged in the plots (Table 2). However, as with total seedlings, fire history, microtopography, and seeding treatment had significant effects on patterns of 2-year recruits (Tables 2 and 4). The binomial model and the count model both indicated that the number of 2-year recruits was lowest in MF sites and in control plots. We consistently found more 2-year recruits in LI sites than in MF or SI sites. No natural 2-year recruits were detected in MF sites, and only two were found in SI sites. Both models showed that the presence of 2-year recruits was significantly more likely in plots that had been seeded. The count model also indicated that 2-year recruits were present significantly more in hollows than on earth hummocks, similar to the total seedlings models. However, as seen in the count model for total seedlings, there was a significant interaction between fire history and microtopography. When 2-year recruits were present in MF sites, more were found on earth hummocks than in hollows.

4. Discussion

Our results indicate that recruitment in short-interval burned stands is limited by seed availability, while recruitment in unburned, mature forest stands is limited by seed bed quality. Post-fire regeneration from natural seed was reduced in mature forest and short-interval burned stands compared to long-interval burns. Once seed was added, regeneration in short-interval stands was equivalent or superior to that in long-interval burned stands. Mature forest stands had little recruitment, even when seed was experimentally added. The contrast between the responses to seeding in short-interval burned sites versus mature forest sites indicates that seedling regeneration in the different stand types is limited by different factors.

These constraints on seedling recruitment are key ecological mechanisms driving fire and vegetation interactions in these forests. Firstly, constraints on seed germination in unburned stands means that cycles of seedling recruitment and stand renewal are primarily restricted to short periods of establishment on seed beds exposed after fire (Johnson and Fryer, 1989; Johnstone et al., 2004). Secondly, and most importantly for predicting the effects of changing disturbance regimes, historical factors that reduce seed stores and post-fire seed availability can dramatically alter patterns of post-fire stand recruitment and dominance (Lamont et al., 1991;

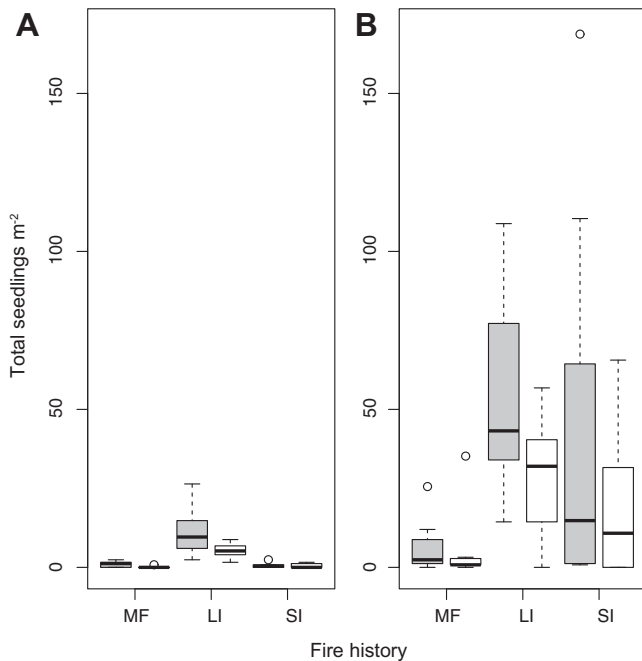


Fig. 2. Total number of black spruce seedlings per m² (a = control, b = seeded plots), grouped by fire history and microtopography. Fire history classes are: MF = mature forest, LI = long-interval burned, and SI = short-interval burned ($n = 8$ sites per fire history class). Grey boxes indicate sites located in hollows within hummocky terrain and white boxes indicate sites located on earth hummocks. Data for seeded plots (b) have been corrected by subtracting the number of black spruce seedlings in control plots in order to show the seeded effects. Boxes encompass the 25–75% quartiles of the data, with the median indicated by the line through the centre of the box. Whiskers extending from the box encompass the 95% quartiles, and extreme observations are shown as hollow circles.

Keeley et al., 1999; Payette et al., 2000; Johnstone and Chapin, 2006b; Johnstone et al., 2010). Because serotinous species depend on onsite seed stores for recruitment (Lamont et al., 1991; Lamont and Enright, 2000), when those stores are limited the feedbacks maintaining the dominance of these species across disturbance intervals are disrupted. In contrast, where seed availability does not limit stand regeneration, e.g., conifers that depend on wind dispersal from adjacent stands for regeneration, a 15-year fire cycle can result in conifer recruitment similar to that of a typical fire return interval (Donato et al., 2009).

Natural experiments such as ours provide valuable opportunities for examining ecological responses to large disturbances that

are difficult to manipulate directly. However, because fire history treatments were not applied as a true experiment, we must consider other plausible hypotheses for a lack of seed and/or natural seedling germination after two closely timed fires. Black spruce could fail to regenerate after a short fire return interval due to poor seed bed quality, i.e., substrate effects. Seed bed quality is known to have important effects on colonisation after fire in the boreal forest (e.g., Jayen et al., 2006; Johnstone and Chapin, 2006a; Greene et al., 2007). However, short-interval burned stands had very low seed availability and very little natural recruitment, thus we can assume with some confidence that regeneration was limited by seed availability. Our results are consistent with other studies of repeat disturbances that suggest low seed availability may be the cause of recruitment failures or reductions of serotinous species after closely timed disturbances (e.g., Landhausser and Wein, 1993; Tsitsoni, 1997; Keeley et al., 1999; Lloret et al., 2003; Jasinski and Payette, 2005; Johnstone and Chapin, 2006b). In this study, data on both seed availability and recruitment under natural and artificially increased seed levels allowed us to conclusively demonstrate the role of seed availability in driving these recruitment failures.

The mature forest stands in our study are more likely candidates for substrate limits on regeneration. These stands have not been burned for more than seventy years (Brown and Johnstone, 2011) and have accumulated thick layers of moss and lichen on the forest floor. Previous research has shown recruitment pulses of conifers associated with seed bed availability after disturbance (Johnson and Fryer, 1989; Johnstone et al., 2004; Peters et al., 2005) and that recruitment is reduced on mature forest floor compared to microsites with higher moisture and nutrient availability, such as decomposing logs (Simard et al., 2003; Peters et al., 2006). Where moss and lichen layers remain intact, seedlings are exposed to large fluctuations in temperature and moisture on the porous seed bed (Charron and Greene, 2002; Johnstone and Chapin, 2006a). If a spruce seed is able to germinate on this type of surface, it is likely to die due to desiccation (Charron and Greene, 2002; Jayen et al., 2006).

The disruption of regeneration pathways reduces the advantage of a serotinous regeneration strategy to favour other strategies, such as long distance dispersal or vegetative resprouting, causing shifts in plant communities (e.g., Johnstone and Chapin, 2006b). We did detect a few black spruce seedlings in short-interval burned stands, thus there is potential for them to provide a seed source once they become reproductively mature, allowing for delayed spruce recruitment. However, our data from mature forest stands shows that conifer recruitment is unlikely once the understory has re-established. Thus there are two constraints on succes-

Table 4

Summary of results from the zero-altered negative binomial model of 2-year old black spruce recruits ($df = 12$). Experimental sites were located in three fire history classes (MF = mature forest, LI = long-interval burned, and SI = short-interval burned; $n = 8$ sites per fire history class), with sites stratified by two microtopography classes (earth hummocks and hollows in hummocky terrain) within each fire history class. Both models included fire history class and seeding treatment as predictor variables. The count model also included microtopography and the interaction between fire history class and microtopography as predictor variables. For seeded plots, the models were based on counts corrected by subtracting counts from control plots within the same site. The intercept represents expected seedling counts in LI stands, hollows, and unseeded control plots. Values in bold indicate that a significant difference ($\alpha \leq 0.05$) was detected in the model.

Model term	Binomial model			Count model		
	Estimate	Standard error	z-value	Estimate	Standard error	z-value
Intercept	−0.5902	0.2132	−2.769	0.2414	0.3482	0.693
SI	−2.2632	0.3031	−7.467	0.6346	0.4414	1.438
MF	−3.0168	0.3550	−8.497	−2.1785	0.5953	−3.659
Hummock ^a	–	–	–	−0.7828	0.3201	−2.445
Seeded ^b	1.8613	0.2793	6.664	1.6732	0.3371	4.963
SI * hummock	–	–	–	−0.4547	0.6451	−0.705
MF * hummock	–	–	–	2.3544	0.8934	2.635

^a Plots located on hummock microtopography.

^b Plots seeded with black spruce in August, 2007.

sion in short-interval burned stands: (1) a lack of available seed for immediate post-fire recruitment; and (2) a seed bed that is inhospitable to spruce seedlings once the few trees that do recruit become reproductively mature. These constraints make serotinous communities vulnerable to rapid and persistent shifts in stand dominance following two closely timed disturbances (Lamont *et al.*, 1991).

Here we used overlapping fires to conduct a natural experiment in which we tested hypotheses related to post-fire recruitment in black spruce forests. Although natural experiments such as this one do not allow us to control many of the confounding factors influencing fire behaviour, conducting an experimental manipulation of landscape-scale fire is not feasible. Thus, we took advantage of a naturally occurring short fire return interval to assess how changes in fire frequency may influence patterns of tree regeneration in these northern boreal forests. The three most recent fires (1990, 1991, and 2005), and presumably the historic fires, were all lightning-ignited and burned during relatively similar conditions (Brown and Johnstone, 2011). The period of the 2005 fire had several individual days of higher Fire Weather Index values than the 1990 and 1991 fire periods, which is reflected in the larger size of that fire. However, the severity of all three recent fires was similar, as indicated by the presence of fine branches on all standing-dead black spruce. Although we cannot know for certain whether the conditions that occurred during the 2005 will become more frequent in the future, they were not so unusual that their reoccurrence would be unlikely.

Because studying naturally occurring burns is inherently opportunistic, we also did not have control over the length of the fire return intervals we studied. The fire history mosaic in which we did our research allowed us to compare short- (14–15 year) and long- (~94-year) interval burned stands adjacent to one another and interspersed with mature forest stands. The results we obtained are consistent with other studies of black spruce that have shown large reductions in seedling recruitment following repeat fires spaced ~25 years apart (Johnstone and Chapin, 2006b) and compound disturbances of spruce budworm outbreak followed by fire after ~30 years (Jasinski and Payette, 2005). Empirical observations of black spruce regeneration after fire from a large-scale study in Alaska indicated that black spruce regeneration was reduced following fire-return intervals of 50–75 years (Johnstone *et al.*, 2010). In addition, modelling simulations based on reproductive life history traits indicate that serotinous conifer systems, such as black spruce-dominated stands, have a reduced ability to recover from fires that occur less than 30 years apart (Suffling, 1995; McIntire *et al.*, 2005). The consistency of these results suggests that the serotinous regeneration strategy of black spruce is sensitive to repeat disturbances occurring with a frequency of less than thirty years, and that this sensitivity may include fire intervals up to 50 years. Given the current range of historical fire intervals on the order of 80–150 years (Viereck, 1983; Larsen, 1997), future climate-driven changes may well result in a large increase in the number of forest stands burning twice within intervals of less than 50 years, with strong consequences for forest regeneration (e.g., Westerling *et al.*, 2011).

The results of our study provide strong empirical evidence of ecological mechanisms by which changes in fire frequency may lead to rapid shifts in stand dominance in the boreal forest. The vegetation of the boreal forest is thought to be highly resilient to climatic change (Chapin *et al.*, 2004), due in part to the adaptations that conifers have that contribute to their self-replacement after fire. Our study provides empirical evidence of black spruce recruitment failure at the northern margin of the boreal forest due to a shortened fire return interval. The indirect climate effects caused by a change to the disturbance regime may initiate vegetation shifts of a larger magnitude or opposite direction than would

happen due to climatic change alone. This shift is counter-intuitive to our current predictions regarding climate warming effects on the northern margin of the boreal forest. Instead of the stand infilling and tree range expansion that is predicted with climatic warming (Kaplan *et al.*, 2003), changes in disturbance regime may initiate the local degradation of treeline forest.

5. Conclusions

Disturbance is a common occurrence in ecological systems, and recovery from disturbance is a key process structuring ecosystems. For plants, recovery success is strongly influenced by two factors: colonisation and establishment. Here we document that a short fire return interval can disrupt strategies selected to ensure the presence of post-fire propagules (serotiny). Northern black spruce forests, such as the one studied here, may be at risk for post-fire regeneration failure with fire return intervals of less than 50 years (Black and Bliss, 1980; Johnstone and Chapin, 2006b; Johnstone *et al.*, 2010). For species adapted to post-disturbance regeneration, substrate conditions may prevent recruitment in undisturbed conditions. This failure to establish after disturbance can cause long-term disruption because the potential for delayed recruitment is limited by the loss of suitable substrates.

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